



Elliptic Curve Cryptography (ECC)

Theory

Elliptic Curve Cryptography (ECC) is a type of public-key cryptography based on the algebraic structure of elliptic curves over finite fields. ECC provides the same level of security as traditional public-key cryptography (like RSA) but with smaller key sizes, making it faster and more efficient.

Elliptic Curve Equation:

An elliptic curve over a finite field F_p is defined as: $y^2 = x^3 + ax + b \pmod{p}$

Where:

- $a, b \in F_p$
- $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$ (to avoid singular curves)
- p is a prime number representing the field

Points on the Curve

A point $P = (x, y)$ satisfies the curve equation. There is a special 'point at infinity', denoted O , which acts as the identity element for addition.

Point Addition

Given two points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ on the curve:

1. Distinct points ($P \neq Q$):

$$m = (y_2 - y_1) / (x_2 - x_1) \pmod{p}$$

2. Point doubling ($P = Q$):

$$m = (3x_1^2 + a) / (2y_1) \pmod{p}$$

3. Resulting point $R = P + Q = (x_r, y_r)$:

$$x_r = m^2 - x_1 - x_2 \pmod{p}$$

$$y_r = m(x_1 - x_r) - y_1 \pmod{p}$$

Scalar Multiplication

Scalar multiplication is the repeated addition of a point: $kP = P + P + \dots + P$ (k times)

Efficient computation uses the double-and-add method, analogous to exponentiation by squaring.

Example

Consider the elliptic curve:

$$y^2 = x^3 + 2x + 3 \pmod{97}$$

and the point $P = (3, 6)$ on the curve.

Point Addition Example

Let $Q = (80, 10)$. To compute $R = P + Q$:

1. Calculate slope:

$$m = (10 - 6) / (80 - 3) \pmod{97} = 4/77 \pmod{97}$$

2. Compute modular inverse of 77 mod 97: $77^{-1} \pmod{97} = 63 \pmod{97}$

2. Compute modular inverse of $12 \bmod 97$: $12^{-1} = 89 \bmod 97$

3. Slope: $m = 4 * 63 \bmod 97 = 252 \bmod 97 = 58$

4. Compute resulting point:

$$x_r = 58^2 - 3 - 80 \bmod 97 = 30$$

$$y_r = 58*(3 - 30) - 6 \bmod 97 = 95$$

So, $P + Q = (30, 95)$.

Scalar Multiplication Example

Compute $3P = P + P + P$:

1. Double P : $2P = P + P$

$$m = (3*3^2 + 2)/(2*6) \bmod 97 = 29/12 \bmod 97$$

Modular inverse of $12 \bmod 97 = 89$

$$m = 29*89 \bmod 97 = 61$$

$$x_{2P} = 61^2 - 3 - 3 \bmod 97 = 2$$

$$y_{2P} = 61*(3 - 2) - 6 \bmod 97 = 55$$

So, $2P = (2, 55)$

2. Add P again: $3P = 2P + P = (2, 55) + (3, 6)$

Procedure:

Colab Notebook Link for this lab: [Colab Lab 7 - ECC](#)